Corey Lynn Murphey

Curriculum Vitae

corey.murphey@colorado.edu
Website | ORCID | Github

EDUCATION

Ph.D. — **University of Colorado – Boulder** — **8/2021 – present**
In Progress
Department of Computer Science
Advisors: Elizabeth Bradley, Allison Hilger, and Jed Brown
Focus: Numerical and Scientific Computing

M.S. — **University of Colorado – Boulder** — **8/2021 – 5/2024**
Department of Computer Science
Advisors: Elizabeth Bradley and Jed Brown
Focus: Numerical and Scientific Computing

M.S. — **Stanford University** — **4/2012 – 4/2014**
Department of Mechanical Engineering
Advisor: Reginald Mitchell
Focus: Energy Systems; Breadth: Biomechanics and Manufacturing

B.S. — **Stanford University** — **8/2008 – 1/2013**
Department of Mechanical Engineering
Advisor: Ellen Kuhl
Focus: Computational Biomechanics and Biomechanical Engineering

RESEARCH EXPERIENCE

6/2021 – present — **University of Colorado – Boulder**
Graduate Research Assistant, Advised by Elizabeth Bradley and Allison Hilger

Developed models of phonation and vibration-induced aerosolization.

Simulated fluid-structure interaction within the larynx.

Generated musical chord progressions using chaotic dynamics and the Tonnetz.

Analyzed Indian ocean sediment core data using Information Theory.

9/2012 – 6/2013 — **Stanford University, Hearing Dynamics**
Research Assistant, Advised by Sunil Puria

Built a model of Békésy's pendulum to demonstrate hair cell dynamics.

Developed a computational model of Békésy's analogy for the inner ear.

5/2010 – 8/2012 — **Stanford University, Living Matter Laboratory**
Research Assistant, Advised by Ellen Kuhl

Modeled electrochemical conductive pathways of the heart.

Generated electrocardiogram plots of simulated cardiac pacing.

Developed patient-specific models of implant-induced skin growth.

Worked with graduate students to create a model of red blood cell division.

Designed a continuum growth model of the vocal folds and vocal polyps.

PUBLICATIONS

2026 **C. Murphey**, A. Hilger, and E. Bradley, "Fluid Atomization from a Clarinet Reed," *In Review*.

2025 A. Hilger, T. Stockman, **C. Murphey**, J. McCurdy, and S. Miller, "The influence of glottal and respiratory factors on aerosol emission during phonation," PLOS ONE, vol. 20, no. 12, e0321909, 2025, doi: 10.1371/journal.pone.0321909.

Reviewed Conference Papers

2011 **C. L. Murphey**, J. Wong, and E. Kuhl, "Computational Simulation of Biventricular Pacing in an Asymptomatic Human Heart," in SBC2011, ASME 2011 Summer Bioengineering Conference, Parts A and B, pp. 917–918, Jun. 2011, doi: 10.1115/SBC2011-53110.

2011 **C. L. Murphey**, J. Wong, and E. Kuhl, "Computational Simulation of Biventricular Pacing in a Human Heart," in Proceedings of the IUTAM Symposium on Computer Models in Biomechanics, Stanford, California, 2011.

Presentations

Talks

2025 **C. L. Murphey** and E. Bradley, "From Climate to Clarinet Reeds: Navigating Flawed Data in Computational Physics," Vassar College Physics Department, Poughkeepsie, NY, Nov. 2025. [*Invited Talk*].

2025 **C. L. Murphey**, E. Bradley, and A. Hilger, "A Computational Model of Phonation-Induced Aerosolization," SIAM Conference on Applications of Dynamical Systems (DS25), Denver, CO, May 2025.

Posters

2025 **C. L. Murphey**, A. Hilger, and E. Bradley, "A Computational Model of Phonation-Induced Aerosolization," Dynamics Days US 2025, Denver, CO, Jan. 2025.

2024 **C. L. Murphey**, A. Hilger, and E. Bradley, "A Dynamics-Inspired Model for Phonation-Induced Aerosolization," University of Colorado – Boulder Computer Science Department's Research Expo Poster Session, Feb. 2024.

2024 Z. Atkins and **C. L. Murphey**, "Generation of Novel Chord Progressions via a Musically-Inspired Chaotic Mapping," University of Colorado – Boulder Computer Science Department's Research Expo Poster Session, Feb. 2024[1].

2024 **C. L. Murphey**, A. Hilger, and E. Bradley, "A Dynamics-Inspired Model for Phonation-Induced Aerosolization," University of Colorado – Boulder Applied Math Department's Research Poster Session, Feb. 2024.

2024 **C. L. Murphey**, A. Hilger, and E. Bradley, "A Dynamics-Inspired Model for Phonation-Induced Aerosolization," Dynamics Days US 2024, Davis, CA, Jan. 2024[2].

[1] Awarded a department award for best work-in-progress poster at the Computer Science Research Expo.

[2] Awarded best poster at Dynamics Days 2024 in Davis, CA.

2024 Z. Atkins and **C. L. Murphey**, "Generation of Novel Chord Progressions via a Musically-Inspired Chaotic Mapping," Dynamics Days US 2024, Davis, CA, Jan. 2024[3].

2023 **C. L. Murphey**, A. Hilger, and E. Bradley, "An Experimentally Validated Model of Phonation-induced Aerosolization," American Association for Aerosol Research 41st Annual Conference (AAAR 2023), Portland, OR, Oct. 2023.

2023 **C. L. Murphey**, A. Hilger, and E. Bradley, "A Dynamics-Inspired Model for Phonation-Induced Aerosolization," SIAM Conference on Applications of Dynamical Systems (DS23), Portland, OR, May 2023.

2023 **C. L. Murphey**, A. Hilger, and E. Bradley, "A Computational Model of Phonation-Induced Aerosolization," 15th International Conference on Advances in Quantitative Laryngology, Voice and Speech Research 2023, Phoenix, AZ, Mar. 2023.

2023 **C. L. Murphey**, A. Hilger, and E. Bradley, "A Dynamics-Inspired Model for Phonation-Induced Aerosolization," University of Colorado – Boulder Applied Math Department's Research Poster Session, Mar. 2023.

2023 **C. L. Murphey**, A. Hilger, and E. Bradley, "A Dynamics-Inspired Model for Phonation-Induced Aerosolization," Dynamics Days US 2023, Virtual, Jan. 2023.

2011 **C. L. Murphey**, J. Wong, and E. Kuhl, "Computational Simulation of Biventricular Pacing in an Asymptomatic Human Heart," ASME Summer Bioengineering Conference, Farmington, PA, Jun. 2011.

Patents

Inventor

2020 US D881,340, "Showerhead and arm," Apr. 14, 2020.

2019 US 10,421,083, "Immersive showerhead," Sep. 24, 2019.

2019 US D855,759, "Shower wand," Aug. 6, 2019.

2018 US 9,931,651, "Immersive showerhead," Apr. 3, 2018.

2018 US 9,925,545, "Immersive showerhead," Mar. 27, 2018.

2018 US D810,233, "Shower wand and adjustable mount," Feb. 13, 2018.

2018 US D810,234, "Showerhead and adjustable bracket," Feb. 13, 2018.

Miscellaneous Publications

2025 **C. L. Murphey**, "Fluctus Cloud," APS Division of Fluid Physics, Traveling Gallery of Fluid Motion, Houston Museum of Natural Science, 2025. [*Photograph*].

2016 G. Parisi-Amon and **C. L. Murphey**, "Full Steam Ahead," ANSYS Advantage, vol. 10, no. 1, pp. 10–12, 2016. [*Magazine Article*].

2013 [*Contributor and Editor*] J. Schox, Not So Obvious: An Introduction to Patent Law and Strategy, 3rd ed. CreateSpace Independent Publishing Platform, 2013. [*Book*].

Software and Code

2021 Time convergence study using SU2 (GitHub)

FUNDING

2026 **NIH Ruth L. Kirschstein National Research Service Award**

[3]Awarded second prize for poster awards at Dynamics Days 2024 in Davis, CA.

(NRSA) Individual Predoctoral Fellowship (F31)
National Institute of Health,
National Institute on Deafness and Other Communication Disorders
Title: "Computational framework for aerosolization from vocal fold vibration"

2019 **Colorado Advanced-Industries Early-Stage Capital and Retention Grant**
State of Colorado, OEDIT

2010 – 2012 **Vice Provost of Undergraduate Education (VPUE) Grant**
Stanford University

2010 – 2012 **Summer Undergraduate Research Institute (SURI) Grant**
Stanford University

2008 **Stanford Summer Engineering Academy (SSEA) Grant**
Stanford University

AWARDS, FELLOWSHIPS, AND HONORS

2024 **Outstanding Departmental Service Award**
University of Colorado – Boulder, Computer Science Department

2024 **Poster Award - Work in Progress Research, Computer Science Research Expo**
University of Colorado – Boulder, Computer Science Department
For the work-in-progress poster entitled "Generation of Novel Chord Progressions via a Musically-Inspired Chaotic Mapping" with primary author, Zachary Atkins.

2024 **First & Second Prize - Poster Awards, Dynamics Days 2024**
For the posters entitled "A Dynamics-Inspired Model for Phonation-Induced Aerosolization" and "Generation of Novel Chord Progressions via a Musically-Inspired Chaotic Mapping".

2023 **D. J. Kasik (1972) Scholarship Fund Award**
University of Colorado – Boulder, College of Engineering and Applied Sciences

2023 **Outstanding Departmental Service Award**
University of Colorado – Boulder, Computer Science Department

2022 **CS Endowed Founder's Fellowship**
University of Colorado – Boulder, Computer Science Department

2021 **Early Career Professional Development Fellowship**
University of Colorado – Boulder, Computer Science Department

Miscellaneous Awards

Current Open water swimming record holder: North Channel (female), Lake Tahoe north-south, and Lake Tahoe double

PROFESSIONAL EXPERIENCE

10/2018 – 7/2021 **Bolder Industries**, Boulder, CO
R&D Engineer, IP Manager, and Chief of Staff

Modeled tire pyrolysis reaction kinetics to anticipate pyrolytic outputs.

5/2018 – 8/2018 **Caban Systems**, San Mateo, CA

Thermal Engineer, Consultant

Modeled heat emitted by batteries inside an energy-storage cabinet.

3/2017 – 4/2018 **Run8 Patent Group**, San Francisco, CA
Patent Agent

Drafted and prosecuted over 50 patent applications.

6/2015 – 3/2017 **Nebia**, San Francisco, CA
R&D Engineer and Engineering Project Manager

Simulated heat-transfer from droplets emitted from showerhead nozzles.

Modeled internal fluid pathways optimized for nozzle performance.

4/2014 – 4/2015 **Schox Patent Group**, San Francisco, CA
Patent Agent

Drafted patent applications and managed portfolios of startups.

6/2013 – 9/2013 **Benvenue Medical Inc.**, Santa Clara, CA
R&D Engineering Intern

Developed surgical protocols for mixing and dispensing bone cement.

CONFERENCES

2026 Dynamics Days US 2026, Tucson, AZ.

2025 SIAM Conference on Applications of Dynamical Systems (DS25), Denver, CO.

2025 Dynamics Days US 2025, Denver, CO.

2024 Dynamics Days US 2024, Davis, CA.

2023 American Association for Aerosol Research (AAAR) 41st Annual Conference, Portland, OR.

2023 SIAM Conference on Applications of Dynamical Systems (DS23), Portland, OR.

2023 15th International Conference on Advances in Quantitative Laryngology, Voice and Speech Research 2023, Phoenix, AZ.

2023 Dynamics Days US 2023, Virtual.

2011 ASME 2011 Summer Bioengineering Conference, Farmington, PA.

2011 IUTAM Symposium on Computer Models in Biomechanics, Stanford, CA.

Conference & Travel Grants

2026 Conference Support Fellowship (CU Boulder Computer Science); Dynamics Days Travel Fellowship

2025 Conference Support Fellowship (CU Boulder Computer Science); Dynamics Days Travel Fellowship

2024 Dynamics Days Travel Fellowship; Clive Bailie Memorial Conference Support Fellowship (CU Boulder)

2023 AAAR US 2023 Student Travel Grant; Conference Support Fellowship (CU Boulder Computer Science); AQL 2023 Student Registration Award; Graduate School Domestic Travel Grant (CU Boulder)

TEACHING

Spring 2026	**Numerical Computation (CSCI 3656)** *Teaching Assistant*, University of Colorado – Boulder
Spring 2026	**Chaotic Dynamics (CSCI 4446/5446)** *Grader*, University of Colorado – Boulder
Summer 2025	**Algorithms (CSCI 3104)** *Teaching Assistant*, University of Colorado – Boulder
Spring 2025	**Numerical Solutions to Partial Differential Equations (CSCI 5636)** *Teaching Assistant*, University of Colorado – Boulder
Spring 2025	**Chaotic Dynamics (CSCI 4446/5446)** *Grader & Course Assistant*, University of Colorado – Boulder
Fall 2024	**Numerical Computation (CSCI 3656)** *Teaching Assistant*, University of Colorado – Boulder
Spring 2023	**Chaotic Dynamics (CSCI 4446/5446)** *Course Manager*, University of Colorado – Boulder
Fall 2013	**Patent Law and Strategy for Engineers (ME 208)** *Course Assistant*, Stanford University
Fall 2012	**Engineering Dynamics (E15)** *Grader*, Stanford University

Guest Lectures

Spring 2026	"Numerical Methods for ODEs," Numerical Computation
Spring 2026	"Gradient Descent," Numerical Computation
Spring 2026	"Convergence of Newton's Method," Numerical Computation
Spring 2026	"Newton's Method," Numerical Computation
Spring 2026	"Automatic and Finite-Difference Differentiation," Numerical Computation
Spring 2026	"Conditioning and Stability of Numerical Methods," Numerical Computation
Summer 2025	"Dynamic Programming," Algorithms
Spring 2025	"Stiffness & Runge-Kutta Methods," Numerical Solutions to PDEs
Fall 2024	"To Grad School or Not Grad School," Advanced Data Structures

SERVICE

Academic Service at the University of Colorado – Boulder

2024 – present	**Computing Advisory Board**[4] *Ph.D. Student Member and Representative*
Spring 2024	**Computer Science Ph.D. Open House** *Graduate Student Organizer*, Computer Science Department
Fall 2023	**Computer Science Graduate Student Association (CSGSA)**

[4] A collaboration between industry partners and the School of Computing to discuss strategic initiatives for University of Colorado computing programs and departments.

CSGSA Chair, Computer Science Department

Spring 2023 **Computer Science Ph.D. Open House**
Graduate Student Organizer and Panelist, Computer Science Department

2022 – 2023 **Computer Science Graduate Committee**
Ph.D. Student Representative, Computer Science Department

Spring 2022 **Computer Science Ph.D. Student Open House**
Graduate Student Panelist, Computer Science Department

Formal Mentoring Programs at the University of Colorado – Boulder

Fall 2024 **Computer Science Ph.D. Application Feedback Program**
Mentor

Fall 2023 **Computer Science Ph.D. Application Feedback Program**
Mentor and Program Organizer, Computer Science Department

Spring 2022 **Summer Program for Undergraduate Research (SPUR)**
Advisor to Mentors, College of Engineering and Applied Sciences

Spring 2022 **Discovery Learning Apprenticeship (DLA) Program**
Mentor and Judge, College of Engineering and Applied Sciences

Peer and Project Mentorship at the University of Colorado – Boulder[5]

2024 – 2025 Wisang Sugiarta, Ph.D. Student, Computer Science Department

2022 – 2023 Zachary Atkins, Ph.D. Student, Computer Science Department
Primary author of two award-winning posters (2024)

2022 – 2023 Maria Valentini, Ph.D. Student, Computer Science Department

2022 – 2023 Aditya Pandey, M.S. Student, Computer Science Department

2022 – 2023 Armin Gholampoor, M.S. Student, Computer Science Department

PROFESSIONAL MEMBERSHIPS, AFFILIATIONS, & CERTIFICATIONS

Certifications

2015 – present United States Patent and Trademark Office, Registered Patent Agent

General Membership

Acoustical Society of America (ASA)

Society for Industrial and Applied Mathematics (SIAM)

The Voice Foundation

Society of Women Engineers (SWE)

American Society of Mechanical Engineers (ASME)

Other Service and Affiliations

2025 – present Boulder High School Women's Swimming : Assistant Coach

2025 – present Boulder Aquatic Masters: Coach

[5]Peer mentorship and project mentorship for fellow graduate students.

2021 – 2023	Renova New Music Ensemble: Founding Member, Webmaster, Soprano/Alto
2021 – 2023	CU – Chamber Singers: Alto 1
2018 – 2020	St. Vrain Innovation Center: Middle School Robotics Mentor